P02

Whole-Genome Exact Pattern Matching and Alignment

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Domain DOM_BIO

Abstract

MUMmer uses suffix trees to identify maximal unique matches between complete bacterial genomes. These anchor points form backbone alignments across extensive chromosomal rearrangements in linear runtime.

Abstract quoted from A. L. Delcher et al., Nucleic Acids Research (1999).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?